JIGAWA STATE INTER-SCHOOL SPELLING BEE CHAMPIONSHIP

System Upgrade — Feature Review Document

Version 2.0

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Organized by:	J-YEDA (Youth Educational Empowerment Association)
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This document is a confidential technical feature review prepared by KowaguruTech for internal CEO review. It outlines all approved system changes requested by the J-YEDA association following the live test session of the Jigawa Spelling Bee Web System.

1. EXECUTIVE SUMMARY

Following the successful live test of the Jigawa Spelling Bee Championship Web System with J-YEDA association members and the KowaguruTech team, a set of critical upgrades were identified and agreed upon. This document captures every requested change in precise technical and operational terms for CEO review and sign-off before development begins.

The upgrade transitions the system from a **fixed-round scoring format** to a **flexible stage-based championship** with advanced features including automatic timer-driven validation, decision locking and reversal, a dedicated tiebreaker dashboard, and real-time audience mark display.

Change Summary at a Glance

Feature	Change Description
Terminology	All 'Rounds' renamed to 'Stages' across every screen and database
Stage Structure	4 stages: 18 → 10 → 5 → 3 students (manual judge-controlled advancement)
Marking System	25 marks per correct word; fresh marks every stage
Attempts	Unlimited per stage; judge sets words-per-student limit per stage
Timer Auto-Validation	Timer expiry triggers automatic Correct/Incorrect based on spelling input
Decision Lock	Correct/Incorrect buttons lock after one click to prevent duplicates
Decision Reversal	Judge can reverse any decision in the current stage from a history panel
Tiebreaker Dashboard	New dedicated page (/tiebreaker) for sudden-death tie-break rounds
Speller Monitor	Updates immediately when 'Next Speller' is clicked — no timer wait
Audience Screen	No premature 'OUT' labels; shows live Attempted + Marks feed
Stage End Flow	System calculates rankings → judge reviews summary → confirms advancement

2. STAGE STRUCTURE & QUALIFICATION LOGIC

The competition is restructured into four distinct stages. Each stage begins with a fresh mark tally (no carry-over from previous stages). Advancement is determined by mark ranking, with the judge manually confirming who advances at the end of each stage.

Stage	Starts With	Result	Advancement Criteria	Word Allocation
Stage 1 (Qualification)	18	10 advance 8 dropped	Top 10 by marks	Each student spells N words (judge sets N)
Stage 2 (Quarter-Final)	10	5 advance 5 dropped	Top 5 by marks	Each student spells N words (judge sets N)
Stage 3 (Semi-Final)	5	3 advance 2 dropped	Top 3 by marks	Each student spells N words (judge sets N)
Stage 4 (Final / Knockout)	3	Champion + 1st & 2nd Runner-up	Sudden Death (miss = out)	Continues until standings are clear

Key Rules

- Marks reset to zero at the start of every stage — previous stage performance has no effect.
- The judge configures the number of words per student per stage in the cockpit settings before or during the stage.
- Attempts per student within a stage are unlimited — the judge manually cycles through students, and the system enforces no automatic cap.
- Students are never marked as eliminated mid-stage. Elimination only occurs after the judge confirms the stage-end summary.
- Stage 4 (Final) operates as Sudden Death — a wrong answer immediately eliminates the student. Final positions (1st, 2nd, 3rd) are determined in this stage.

3. MARKING SYSTEM

Event	Mark Awarded
Correct spelling	25 marks
Incorrect spelling	0 marks
Auto-marked (timer expired, correct)	25 marks
Auto-marked (timer expired, incorrect)	0 marks
Reversed decision (Correct → Incorrect)	-25 marks from student total
Reversed decision (Incorrect → Correct)	+25 marks to student total

The system replaces the previous 10-points-per-word model. All four stages use 25 marks per correct word. The Final stage (Sudden Death) still records marks for transparency but advancement is based purely on whether the student spells correctly or not.

4. TIMER AUTO-VALIDATION

When the countdown timer reaches zero while a student is actively spelling, the system automatically evaluates the spelling entered and records the result — no judge button click required.

Step	Action	Who
1	Timer reaches 0:00	System (automatic)
2	System compares typed letters with the target word (case-insensitive, punctuation stripped)	System (automatic)
3	If match → fires MARK CORRECT (25 marks)	System (automatic)
4	If no match → fires MARK INCORRECT (0 marks)	System (automatic)
5	Result displayed on all screens instantly	System (automatic)
6	Judge clicks 'Next Speller' to continue	Judge

Note: If the judge has already clicked Correct or Incorrect before the timer reaches zero, the auto-validation does not fire — the manual decision takes priority.

5. DECISION LOCK & REVERSAL SYSTEM

5.1 One-Click Lock

To prevent accidental double-clicks and duplicate scores, the Correct and Incorrect buttons on the judge cockpit are locked immediately after a single click.

- Both buttons are greyed out and disabled the moment one is clicked.

- A visual badge shows the recorded result (e.g. '✔ Correct — 25 marks recorded').
- The lock is cleared automatically when the judge clicks 'Next Speller'.
- Auto-validation (timer expiry) also triggers the lock.

5.2 Decision Reversal Panel

A collapsible 'Reversal / Decision History' panel is available on the judge cockpit. This panel allows the judge to review and correct any decision made within the current stage.

- Displays a full chronological list of all decisions in the current stage.
- Each entry shows: Student Name, Word, Decision (Correct / Incorrect), Marks, Time.
- Each entry has a '■ Reverse' button.
- Reversing a decision: removes the history record, adjusts the student's mark total, and immediately broadcasts the updated state to all screens.
- Any decision in the current stage can be reversed — not limited to the most recent.
- Reversed decisions are logged for audit purposes.

6. STAGE END FLOW — MANUAL WITH CONFIRMATION

The judge controls when a stage ends. There is no automatic end trigger. The flow is designed to give the judge full control and visibility before any students are advanced or eliminated.

Step	Action
1	Judge clicks 'End Stage' button in the cockpit
2	System calculates mark rankings for all students in the current stage
3	A Stage Summary Modal appears — showing a full ranked table with colour coding: ● Green = Advances to next stage ● Red = Dropped / Eliminated ● Orange = Tied at cutoff (tiebreaker required)
4	If a tie exists at the cutoff point, a '■ Tiebreaker Required' alert is shown with a button to open the Tiebreaker Dashboard
5	Judge reviews the summary and clicks 'Confirm & Advance'
6	System marks dropped students as eliminated, resets advancing students' marks to 0, moves competition to the next stage, and broadcasts to all screens

7. TIEBREAKER / ADDITIONAL KNOCKOUT DASHBOARD

When two or more students are tied at the qualification cutoff at the end of a stage, a dedicated Tiebreaker Dashboard is used to resolve the tie before advancement is confirmed.

Item	Detail
Access	New button in judge cockpit navigation bar: '■ Open Tiebreaker' (alongside Open Projector, Open Speller, Open Leaderboard)
Page URL	/tiebreaker — a dedicated full-screen page
Word Bank	Reserved words Q401–Q500 are used exclusively for tiebreakers
Format	Sudden Death Knockout — miss a word → immediately eliminated from the tie
Students	Judges select which tied students participate in the tiebreaker
Outcome	Last student(s) remaining fill the available advancement slots; result feeds back automatically into the main co
Display	Has its own timer, pronouncer card, and Correct/Incorrect buttons — identical in layout to the main judge cock

8. AUDIENCE / PROJECTOR SCREEN IMPROVEMENTS

8.1 No Premature Elimination Labels

Previously, a student was shown as 'OUT' immediately after missing a single word. This is removed. No student is labelled as eliminated until the judge has confirmed the stage-end summary.

- During an active stage: all participating students show status as 'Competing'.
- After stage confirmation: dropped students show 'Stage X — Eliminated'.
- Advanced students show 'Advanced to Stage X+1'.

8.2 Live Attempted & Marks Feed

A live-updating results feed replaces the previous static round-history panel. This feed shows the audience each student's running performance within the current stage.

Student Name	School	Words Attempted	Marks	Status
Humaira Sani	MBPS Gumel	3	75	Competing
Ahmad Usman	Mega School Dutse	2	50	Competing
Zainab Yahaya	Mega School Dutse	2	25	Competing
Musa Yakubu	MBPS Bulangu	1	0	Competing
...	...	...	...	...

Example layout of the live Attempted & Marks feed visible on the audience projector screen.

9. SPELLER MONITOR — IMMEDIATE NEXT SPELLER UPDATE

The Speller Monitor screen (displayed downstage for the contestant to see) currently only updates when the judge starts the timer. This creates a gap where the screen shows the previous contestant's information.

	Current Behaviour	New Behaviour (v2)
Trigger for update	Timer Start button clicked	'Next Speller' button clicked
What audience sees	Previous contestant's info until timer starts	New contestant's name, school, and stage immediately
Timer display	Starts counting when judge clicks Start	Shows timer in 'Ready' state (full duration, not ticking) immediately; starts counting when judge clicks Start

10. TECHNICAL — FILES TO BE MODIFIED

File	Changes
db/schema.sql	Add decision_locked, stage_word_limit columns; rename round → stage
db/index.js	Update getFullState(); STAGE_RULES; 25 pts/word
server.js	New endpoints: /api/stage/end, /api/stage/confirm-advance, /api/decision/reverse. Auto-timer validation. Decision lock logic.
public/judge.html	Stage labels; words-per-student input; lock buttons; reversal panel; End Stage button; confirmation modal; Tiebreaker nav button
public/projector.html	Attempted+Marks feed; no mid-stage OUT labels; Stage labels
public/speller.html	Immediate update on Next Speller click
public/leaderboard.html	Stage labels
public/tiebreaker.html	NEW PAGE — Complete Tiebreaker Knockout Dashboard

Implementation Order

- 1. DB Schema — add new columns (no breaking changes to existing data)
- 2. db/index.js — update getFullState(), marks, stage rules
- 3. server.js — new endpoints, timer auto-validation, decision lock
- 4. public/judge.html — UI updates (stages, word-limit, lock, reversal, End Stage modal)
- 5. public/speller.html — immediate Next Speller update
- 6. public/projector.html — Attempted+Marks feed, stage labels
- 7. public/tiebreaker.html — New page
- 8. public/leaderboard.html — Stage label updates
- 9. End-to-end test of full competition flow

11. REVIEW & APPROVAL

Please review all sections of this document and confirm your approval before development begins. All features listed are based on the test session feedback from J-YEDA and the KowaguruTech team.

Name	Role	Signature	Date
	KowaguruTech CEO		
	J-YEDA Representative		
	KowaguruTech Lead Engineer		